

Steering Column



Special Tool(s)	
 ST2574-A	Flex Probe Kit NUD105-R025D or equivalent
 ST3093-A	Fluke 77-IV Digital Multimeter FLU77-4 or equivalent
 ST3094-A	Test Light SGT 27000 or equivalent 250-300 mA incandescent bulb test lamp
 ST2834-A	Vehicle Communication Module (VCM) and Integrated Diagnostic System (IDS) software with appropriate hardware, or equivalent scan tool

Principles of Operation

Power Adjustable Tilt/Telescope Steering Column

The power adjustable tilt/telescope steering column is controlled by the steering column control switch and 2 electric motors that are powered by the Steering Column Control Module (SCCM) . The steering column control switch is a 4-position, momentary-contact switch. When the steering column control switch is moved to any of the 4 positions, it sends a signal to the SCCM to tilt the steering column up or down or telescope the steering column forward or rearward.

The SCCM continuously monitors the steering column tilt and telescoping motors and circuits for faults. If a fault is detected, the SCCM is capable of setting DTCs.

Tilt/Telescoping Motor Sensors

The SCCM receives digital signals from Hall-effect type rotation sensors that are integral to the tilt and telescoping motors. The SCCM uses these signals to calculate the location of the steering column in relation to the full forward/rearward and full up/down positions. The SCCM also uses these signals to store and recall memory positions, carry out the easy entry/exit operations and to make sure the tilt and telescoping motors do not stall against the steering column end of travel in the forward/rearward or up/down directions. Both the tilt and telescopic motors can be serviced separately from the steering column.

Jog Mode

If the **SCCM** loses the signal from either of the motor sensors, the affected steering column motor operates in jog mode. Jog mode allows limited operation of the affected steering column motor using only the steering column control switch. When the steering column control switch is operated in jog mode, the steering column moves in the desired direction for one second, then stops. The steering column control switch must be released, then pressed again in order to move the steering column for an additional second. Jog mode is an indication that there is a tilt or telescoping motor sensor fault. If the adjustable steering column is operating in jog mode, a DTC can set in the **SCCM** .

Memory Position Programming and Recall

The **SCCM** monitors steering column position using Hall-effect sensors that are integral to the tilt and telescoping motors. When a memory position is recalled through either the memory SET switch or the Remote Keyless Entry (RKE) transmitter, the **SCCM** monitors the Hall-effect sensors and stops sending voltage to the steering column motors once the commanded memory position is reached. If the **SCCM** receives a steering column switch input during a memory position recall function, the module stops the memory recall and responds to the new steering column switch position input.

When the **RKE** transmitter is activated, a message is sent over the Medium Speed Controller Area Network (MS-CAN) to the Driver Seat Module (DSM) . When the memory SET switch is pressed, the **DSM** senses continuity to ground on the switch circuit. When the **DSM** receives either the **MS-CAN** message or the ground input through the switch circuit, it sends a message over the **MS-CAN** to the Body Control Module (BCM) . The **BCM** acts as a gateway module, relaying the **MS-CAN** message that was received from the **DSM** to the **SCCM** over the High Speed Controller Area Network (HS-CAN) . The **SCCM** then sends voltage over the appropriate circuit to the steering column tilt or telescoping motor to adjust the steering column position.

For information on setting and recalling a memory position, refer to the Owner's Literature or Memory Position Programming in Section 501-10.

Easy Entry/Exit

The easy entry/exit function moves the steering column to the full inward and upward position when the selector lever is in the PARK position and the ignition switch indicates the ignition key has been removed from the ignition lock cylinder. The **SCCM** receives an ignition status message over the High Speed Controller Area Network (HS-CAN) . When the ignition status message indicates the ignition key has been removed from the lock cylinder, the **SCCM** commands the steering column to the full upward/forward position. The **SCCM** cancels this operation if a valid input is received from the steering column control switch or memory position switch.

The **SCCM** records the current steering column position before powering the steering column for an easy exit operation. This recorded position returns the steering column to this position on the next easy entry operation. During easy entry operation, the steering column is returned to the position previous to the easy exit operation.

The easy entry/exit feature can be activated or deactivated using the message center. Refer to driver controls in the Owner's Literature for additional instructions for using the message center.

Power Tilt Steering Column Hard Stop/Soft Stop

A hard stop occurs when the power column physically reaches the end of travel and cannot go any further. A soft stop occurs when the power column stops before physically reaching the end of travel. The hard stop is set by steering column design and cannot be changed or adjusted. The soft stop is set by the **SCCM** . To prevent unnecessary stress on the power column and motors, the **SCCM** sets 4 soft stop positions, one for tilt up, one for tilt down, one for telescope out and one for telescope in. The **SCCM** uses a preset distance from the hard stop to determine where the soft stop occurs.

The soft stops must be relearned by the **SCCM** whenever a new steering column or **SCCM** is installed.

Inspection and Verification

1. Verify the customer concern.
2. Visually inspect for obvious signs of mechanical or electrical damage.
3. Visually inspect for obstructions preventing column movement.

Visual Inspection Chart

Mechanical	Electrical
- Steering column control switch - Adjustable steering column tilt motor - Adjustable steering column telescope motor - Steering column assembly	- Wiring, terminals or connectors - Steering Column Control Module (SCCM)

4. If an obvious cause for an observed or reported concern is found, correct the cause (if possible) before proceeding to the next step.

5. **NOTE:** *Make sure to use the latest scan tool software release.*

If the cause is not visually evident, connect the scan tool to the Data Link Connector (DLC) .

6. **NOTE:** *The Vehicle Communication Module (VCM) LED prove out confirms power and ground from the DLC are provided to the VCM .*

If the scan tool does not communicate with the VCM :

- check the VCM connection to the vehicle.
- check the scan tool connection to the VCM .
- refer to Section 418-00, No Power To The Scan Tool, to diagnose no power to the scan tool.

7. If the scan tool does not communicate with the vehicle:

- verify the ignition is ON.
- verify the scan tool operation with a known good vehicle.
- refer to Section 418-00, The PCM Does Not Respond To The Scan Tool, to diagnose no response from the PCM.

8. Carry out the network test.

- If the scan tool responds with no communication for one or more modules, refer to Section 418-00.
- If the network test passes, proceed with module self-test diagnostics.

9. Carry out the self-test diagnostics for the SCCM .

10. If the DTCs retrieved are related to the concern, go to Steering Column Control Module (SCCM) DTC Chart. For all other SCCM DTCs, refer to the Master DTC Chart in Section 211-05.

11. If no DTCs related to the concern are retrieved, GO to [Symptom Chart](#).

DTC Chart

Steering Column Control Module (SCCM) DTC Chart

DTC	Description	Action
B1298:09	Steering Column Adjust Up Switch: Component Failure	INSTALL a new steering column control switch. REFER to Brake Pedal Control Switch in Section 206-06. CLEAR the DTCs. VERIFY correct system operation. REPEAT the self-test.

DTC	Description	Action
		If DTC B1298:09 returns, INSTALL a new Steering Column Control Module (SCCM) . REFER to Section 211-05. CLEAR the DTCs. VERIFY correct system operation. REPEAT the self-test.
B1299:09	Steering Column Adjust Down Switch: Component Failure	INSTALL a new steering column control switch. REFER to Brake Pedal Control Switch in Section 206-06. CLEAR the DTCs. VERIFY correct system operation. REPEAT the self-test. If DTC B1299:09 returns, INSTALL a new SCCM . REFER to Section 211-05. CLEAR the DTCs. VERIFY correct system operation. REPEAT the self-test.
B12A1:09	Steering Column Adjust Out Switch: Component Failure	INSTALL a new steering column control switch. REFER to Brake Pedal Control Switch in Section 206-06. CLEAR the DTCs. VERIFY correct system operation. REPEAT the self-test. If DTC B12A1:09 returns, INSTALL a new SCCM . REFER to Section 211-05. CLEAR the DTCs. VERIFY correct system operation. REPEAT the self-test.
B12A2:09	Steering Column Adjust In Switch: Component Failure	INSTALL a new steering column control switch. REFER to Brake Pedal Control Switch in Section 206-06. CLEAR the DTCs. VERIFY correct system operation. REPEAT the self-test. If DTC B12A2:09 returns, INSTALL a new SCCM . REFER to Section 211-05. CLEAR the DTCs. VERIFY correct system operation. REPEAT the self-test.
B1375:18	Steering Column Tilt Motor: Circuit Current Below Threshold	GO to Pinpoint Test A.
B1375:19	Steering Column Tilt Motor: Circuit Current Above Threshold	GO to Pinpoint Test A.
B1375:71	Steering Column Tilt Motor: Actuator Stuck	GO to Pinpoint Test A.
B137E:18	Steering Column Telescope Motor: Current Below Threshold	GO to Pinpoint Test A.
B137E:19	Steering Column Telescope Motor: Current Above Threshold	GO to Pinpoint Test A.
B137E:71	Steering Column Telescope Motor: Actuator Stuck	GO to Pinpoint Test A.
B1C33:11	Steering Column Tilt Feedback Signal: Circuit Short To Ground	GO to Pinpoint Test C.

DTC	Description	Action
B1C33:12	Steering Column Tilt Feedback Signal: Circuit Short To Battery	GO to Pinpoint Test C.
B1C35:11	Steering Column Telescope Feedback Signal: Circuit Short To Ground	GO to Pinpoint Test C.
B1C35:12	Steering Column Telescope Feedback Signal: Circuit Short To Battery	GO to Pinpoint Test C.
—	All Other SCCM DTCs	REFER to Section 211-05.

Symptom Chart

Symptom Chart		
Condition	Possible Causes	Action
No communication with the Steering Column Control Module (SCCM)	- Fuse - Wiring, terminals or connectors - SCCM	REFER to Section 418-00 to diagnose the no communication condition.
The adjustable steering column is inoperative in one or more directions — using the steering column control switch	- Fuse - Wiring, terminals or connectors - Steering column control switch - Adjustable steering column tilt motor - Adjustable steering column telescopic motor - SCCM	GO to Pinpoint Test A.
The adjustable steering column is inoperative in one or more directions — using the memory SET switch buttons	- Wiring, terminals or connectors - Memory SET switch - Driver Seat Module (DSM) - SCCM - Body Control Module (BCM)	GO to Pinpoint Test B.

The adjustable steering column is inoperative in one or more directions — using an Remote Keyless Entry (RKE) transmitter	- Wiring, terminals or connectors RKE transmitter DSM SCCM BCM	GO to Pinpoint Test B.
The adjustable steering column is inoperative in one or more directions — using the easy entry/exit feature	- Entry/exit feature is disabled in the message center - Wiring, terminals or connectors SCCM	GO to Pinpoint Test B.
The adjustable steering column operates in one second intervals when using the steering column control switch (jog mode)	- Wiring, terminals or connectors - Adjustable steering column tilt motor - Adjustable steering column telescopic motor SCCM	GO to Pinpoint Test C.
Looseness felt in the steering wheel or tilt mechanism	- Loose tilt motor fastener(s) - Loose steering column assembly fastener(s) - Wore internal steering column component(s) or loose internal fastener/fasteners - Steering column tilt housing fastener(s)	- TIGHTEN fasteners. For additional information, refer to Steering Column Motor — Tilt or Steering Column Motor — Telescopic. - INSTALL a new steering column. For additional information, REFER to Steering Column in this section.

Pinpoint Tests

Pinpoint Test A: The Adjustable Steering Column is Inoperative in One or More Directions — Using the Steering Column Control Switch

Refer to Wiring Diagrams Cell [128](#) , Adjustable Steering Column for schematic and connector information.

Normal Operation

When the steering column control switch is activated, the Steering Column Control Module (SCCM) supplies voltage and ground to the applicable steering column tilt or telescoping motor to move the steering column to the desired position.

This pinpoint test is intended to diagnose the following:

- Fuse
- Wiring, terminals or connectors
- Steering column control switch
- Adjustable steering column tilt motor
- Adjustable steering column telescopic motor
- SCCM

DTC Description	Fault Trigger Conditions
• B1375:18 — Steering Column Tilt Motor: Circuit Current Below Threshold	This DTC sets when the SCCM senses current flow to the adjustable steering column tilt motor is below a preset limit while the steering column control switch is activated. This may be caused by an open or high resistance in the adjustable steering column tilt motor circuits.
• B1375:19 — Steering Column Tilt Motor: Circuit Current Above Threshold	This DTC sets when the SCCM senses current flow to the adjustable steering column tilt motor is above a preset limit while the steering column control switch is activated. This may be caused by a short to ground in the adjustable steering column tilt motor circuits.
• B1375:71 — Steering Column Tilt Motor: Actuator Stuck	This DTC sets when the SCCM senses current flow to the adjustable steering column tilt motor is within a specific range but above a preset limit during self test.
• B137E:18 — Steering Column Telescope Motor: Current Below Threshold	This DTC sets when the SCCM senses current flow to the adjustable steering column telescope motor is below a preset limit while the steering column control switch is activated. This may be caused by an open or high resistance in the adjustable steering column telescope motor circuits.
• B137E:19 — Steering Column Telescope Motor: Current Above Threshold	This DTC sets when the SCCM senses current flow to the adjustable steering column telescope motor is above a preset limit while the steering column control switch is activated. This may be caused by a short to ground in the adjustable steering column telescope motor circuits.
• B137E:71 — Steering Column Telescope Motor: Actuator Stuck	This DTC sets when the SCCM senses current flow to the adjustable steering column telescope motor is within a specific range but above a preset limit during on-demand self test.

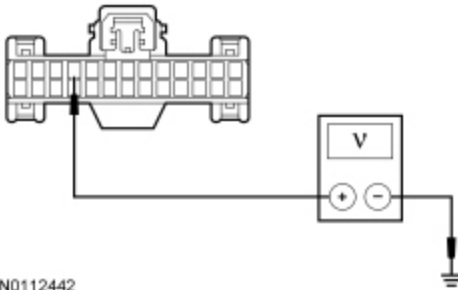
PINPOINT TEST A : THE ADJUSTABLE STEERING COLUMN IS INOPERATIVE IN ONE OR MORE DIRECTIONS — USING THE STEERING COLUMN CONTROL SWITCH

NOTICE: Use the Flex Probe Kit for all test connections to prevent damage to the wiring terminals. Do not use standard multi-meter probes.	
NOTE: If the adjustable steering column only operates in one second intervals when activating the steering column control switch, GO to Pinpoint Test C.	
A1 CHECK THE OPERATION OF THE ADJUSTABLE COLUMN USING THE STEERING COLUMN CONTROL SWITCH	
• Operate the adjustable steering column in all directions using the steering column control switch.	
Does the steering column operate correctly in at least one direction using the steering column control switch?	
Yes	GO to A4.

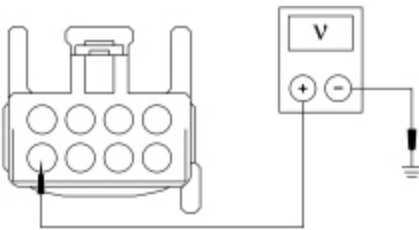
No	If the steering column does not move in any direction at all using the steering column switch, GO to A2. If the steering column moves in only one second intervals (jog mode) in one or more directions using the steering column switch, GO to Pinpoint Test C.
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A2 CHECK FOR VOLTAGE AT THE SCCM

- Ignition OFF.
- Disconnect: SCCM [C2414A](#) .
- Disconnect: SCCM [C2414B](#) .
- Measure the voltage between SCCM [C2414A](#) Pin 10, circuit SBP23 (WH/RD), harness side and ground.



- Measure the voltage between SCCM [C2414B](#) Pin 8, circuit SBP24 (VT/RD), harness side and ground.

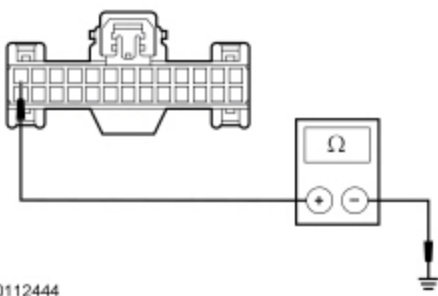


Are the voltages greater than 10 volts?

Yes	GO to A3 .
No	VERIFY BCM fuses 23 (15A) and 24 (15A) are OK. If OK, REPAIR the affected circuit(s). If not OK, REFER to the Wiring Diagrams manual to identify the cause for the circuit short. TEST the system for normal operation.

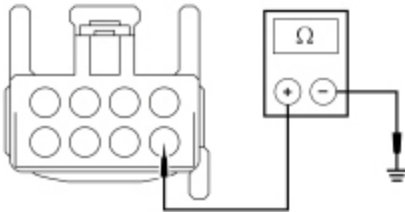
A3 CHECK THE SCCM GROUND CIRCUITS FOR AN OPEN

- Measure the resistance between SCCM [C2414A](#) Pin 13, circuit GD133 (BK), harness side and ground.



• N0112444

- Measure the resistance between SCCM [C2414B](#) Pin 5, circuit GD138 (BK/WH), harness side and ground.



• N0119446

Are the resistances less than 5 ohms?

Yes	GO to A4 .
No	REPAIR the affected circuit(s). TEST the system for normal operation.

A4 CHECK THE SCCM STEERING COLUMN ADJUST SWITCH (STRCOL_ADJ) PID

- Connect: SCCM [C2414A](#) (if previously disconnected).
- Connect: SCCM [C2414B](#) (if previously disconnected).
- Ignition ON.
- Enter the following diagnostic mode on the scan tool: DataLogger — SCCM .
- Monitor the STRCOL_ADJ PID while moving the steering column control switch in all directions.

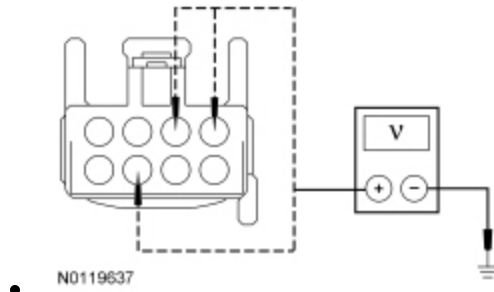
Do the PID values agree with the steering column control switch positions?

Yes	GO to A5 .
No	INSTALL a new steering column control switch. REFER to Brake Pedal Control Switch in Section 206-06. CLEAR the DTCs. VERIFY correct system operation. REPEAT the self-test If the concern is still present, INSTALL a new Steering Column Control Module (SCCM) . REFER to Section 211-05. TEST the system for normal operation.

A5 CHECK THE ADJUSTABLE STEERING COLUMN MOTOR CIRCUITS FOR A SHORT TO VOLTAGE

- Ignition OFF.

- Disconnect: SCCM [C2414B](#) .
- Disconnect: SCCM [C2414C](#) .
- Ignition ON.
- Measure the voltage between ground and:
 - SCCM [C2414B](#) Pin 1, circuit CPP22 (GY/BN), harness side.
 - SCCM [C2414B](#) Pin 2, circuit CPP04 (GY), harness side.
 - SCCM [C2414B](#) Pin 7, circuit CPP02 (BN), harness side.

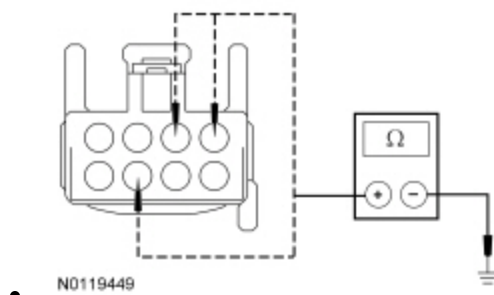


Is any voltage present?

Yes	REPAIR the affected circuit(s). TEST the system for normal operation.
No	GO to A6 .

A6 CHECK THE ADJUSTABLE STEERING COLUMN MOTOR CIRCUITS FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the resistance between ground and:
 - SCCM [C2414B](#) Pin 1, circuit CPP22 (GY/BN), harness side.
 - SCCM [C2414B](#) Pin 2, circuit CPP04 (GY), harness side.
 - SCCM [C2414B](#) Pin 7, circuit CPP02 (BN), harness side.



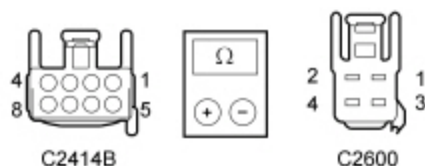
Are the resistances greater than 10,000 ohms?

Yes	GO to A7 .
No	REPAIR the affected circuit(s). TEST the system for normal operation.

A7 CHECK THE ADJUSTABLE STEERING COLUMN MOTOR CIRCUITS FOR AN OPEN

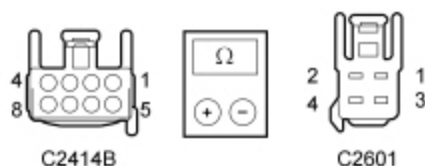
- Disconnect: Adjustable Steering Column Tilt Motor [C2600](#) or Adjustable Steering Column Telescope Motor [C2601](#).
- For an inoperative adjustable steering column tilt motor, measure the resistance between SCCM [C2414B](#) and adjustable steering column tilt motor [C2600](#) using the following chart:

SCCM	Circuit	Adjustable Steering Column Tilt Motor
C2414B Pin 7	CPP02 (BN)	C2600 Pin 2
C2414B Pin 2	CPP04 (GY)	C2600 Pin 1



- N0122534
- For an inoperative adjustable steering column telescope motor, measure the resistance between SCCM [C2414B](#) and adjustable steering column telescope motor [C2601](#) using the following chart:

SCCM	Circuit	Adjustable Steering Column Telescope Motor
C2414B Pin 7	CPP02 (BN)	C2601 Pin 2
C2414B Pin 1	CPP22 (GY/BN)	C2601 Pin 1



- N0122581

Are the resistances less than 5 ohms?

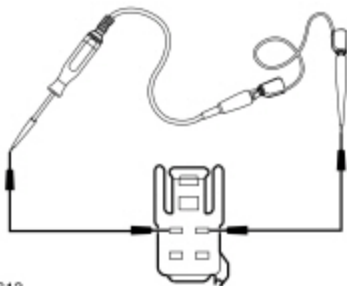
Yes	For a tilt motor that is inoperative/does not operate correctly, GO to A8 . For a telescope motor that is inoperative/does not operate correctly, GO to A9 .
No	REPAIR the affected circuit(s). CLEAR the DTCs. REPEAT the self-test.

A8 CHECK THE SCCM OUTPUT TO THE ADJUSTABLE STEERING COLUMN TILT MOTOR

- Ignition OFF.

- Connect: SCCM [C2414B](#) .
- Connect: SCCM [C2414C](#) .
- **NOTICE:** The following step uses a test light to simulate normal circuit loads. Use only the test light recommended in the Special Tools table at the beginning of this section. To avoid connector terminal damage, use the Flex Probe Kit for the test light probe connection to the vehicle. Do not use the test light probe directly on any connector.

Connect a 12-volt incandescent test light between adjustable steering column tilt motor [C2600](#) Pin 1, circuit CPP02 (BN), harness side and [C2600](#) Pin 2, circuit CPP04 (GY), harness side.



- N0122613
- Ignition ON.
- **NOTE:** Because the steering column motor (Hall-effect sensor) is disconnected, the SCCM only activates the output for approximately 1 second each time the steering column control switch is activated (jog mode).

Activate the steering column control switch in both the UP and DOWN positions (release the switch briefly between each switch activation) while observing the test light.

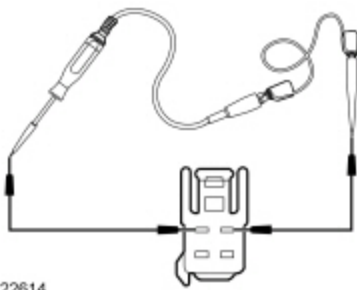
Does the test light momentarily illuminate when the steering column control switch is activated in both UP and DOWN positions?

Yes	INSTALL a new adjustable steering column tilt motor. REFER to Steering Column Motor — Tilt in this section. CLEAR the DTCs. TEST the system for normal operation. REPEAT the self test.
No	GO to A10 .

A9 CHECK THE SCCM OUTPUT TO THE ADJUSTABLE STEERING COLUMN TELESCOPE MOTOR

- Ignition OFF.
- Connect: SCCM [C2414B](#) .
- Connect: SCCM [C2414C](#) .
- **NOTICE:** The following step uses a test light to simulate normal circuit loads. Use only the test light recommended in the Special Tools table at the beginning of this section. To avoid connector terminal damage, use the Flex Probe Kit for the test light probe connection to the vehicle. Do not use the test light probe directly on any connector.

Connect a 12-volt incandescent test light between adjustable steering column telescope motor [C2601](#) Pin 2, circuit CPP02 (BN), harness side and [C2601](#) Pin 1, circuit CPP22 (GY/BN), harness side.



• N0122614

- Ignition ON.
- **NOTE:** Because the steering column motor (Hall-effect sensor) is disconnected, the SCCM only activates the output for approximately 1 second each time the steering column control switch is activated (jog mode).

Activate the steering column control switch in both the FORWARD and REARWARD positions (release the switch briefly between each switch activation) while observing the test light.

Does the test light momentarily illuminate when the steering column control switch is activated in both FORWARD and REARWARD positions?

Yes	INSTALL a new adjustable steering column telescopic motor. REFER to Steering Column Motor — Telescopic in this section. CLEAR the DTCs. TEST the system for normal operation. REPEAT the self test.
No	GO to A10 .

A10 CHECK THE SCCM CONNECTORS

- Check the SCCM electrical connectors for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect: Reconnect all previously disconnected electrical connectors.
- Make sure the connectors are seated correctly then operate the system and verify the concern is still present.

Is the concern still present?

Yes	INSTALL a new SCCM . REFER to Section 211-05. TEST the system for normal operation.
No	The system is operating correctly at this time. Concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

Pinpoint Test B: The Adjustable Steering Column is Inoperative in One or More Directions — Using the Memory SET Switch Buttons/Using an RKE Transmitter/Using the Easy Entry/Exit Feature

Refer to Wiring Diagrams Cell [128](#) , Adjustable Steering Column for schematic and connector information.

Normal Operation

When the desired memory position is selected using the memory SET switch or an Remote Keyless Entry (RKE) transmitter, the Driver Seat Module (DSM) communicates the desired steering column position in a message on the Medium Speed Controller Area Network (MS-CAN) to the Body Control Module (BCM) . The BCM acts as a gateway module, relaying the message to the Steering Column Control Module (SCCM) over the High Speed Controller Area Network (HS-CAN) . The SCCM monitors the position of the adjustable steering column using Hall-effect type rotation sensors that are integral to the tilt and telescoping motors. The sensors provide a digital signal used by the SCCM to calculate the position of the steering column in relation to the full forward/rearward and full up/down positions. The SCCM uses these signals to store and recall memory positions, carry out easy entry/exit operations and to make sure the tilt and telescope motors do not stall against the steering column end of travel. If a tilt or telescope motor sensor DTC sets, the SCCM does not recall a programmed memory position or automatically position the steering column during the easy entry/exit operation. If a DTC sets for either sensor, the affected adjustable steering column motor operates in jog mode. Jog mode allows limited operation of the affected steering column motor using only the steering column control switch. When the steering column control switch is operated in jog mode, the steering column moves in the desired direction for one second, then stops. The steering column control switch must be released, then pressed again in order to move the steering column for an additional second.

This pinpoint test is intended to diagnose the following:

- Entry/exit feature is disabled in the message center
- Wiring, terminals or connectors
- Memory SET switch
- RKE transmitter
- DSM
- BCM
- SCCM

PINPOINT TEST B : THE ADJUSTABLE STEERING COLUMN IS INOPERATIVE IN ONE OR MORE DIRECTIONS — USING THE MEMORY SET SWITCH BUTTONS/USING AN RKE TRANSMITTER/USING THE EASY ENTRY/EXIT FEATURE

NOTICE: Use the Flex Probe Kit for all test connections to prevent damage to the wiring terminals. Do not use standard multi-meter probes.

B1 CHECK THE OPERATION OF THE ADJUSTABLE COLUMN USING THE STEERING COLUMN CONTROL SWITCH

- Operate the adjustable steering column in all directions using the steering column control switch.

Does the steering column move for more than one second in each direction using the steering column control switch?

Yes	If the adjustable steering column is inoperative using the easy entry/exit feature, GO to B2 . If the adjustable steering column is inoperative using the memory SET switch, GO to B3 . If the adjustable steering column is inoperative using an RKE transmitter, GO to B4 .
No	If the steering column does not move in any direction at all using the steering column switch, GO to Pinpoint Test A . If the steering column moves in only one second intervals (jog mode) in one or more directions using the steering column switch, GO to Pinpoint Test C .

B2 VERIFY THE EASY ENTRY/EXIT FEATURE IS ENABLED

- Verify the easy entry/exit feature is enabled using the message center. Refer to the Owner's Literature for additional instructions for using the message center.

Does the message center indicate that the easy entry/exit feature is enabled?

Yes	GO to B5 .
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No	TOGGLE the easy entry/exit feature on using the message center controls. TEST the system for normal operation.
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B3 CHECK THE OPERATION OF THE MEMORY SEATS, MEMORY MIRRORS AND STEERING COLUMN USING THE MEMORY SET SWITCH

- Check the operation of the seats, memory mirrors and steering column using the memory SET switch. Refer to the Owner's Literature for additional instructions for programming a memory position.

Do the seats and mirrors operate properly using the memory SET switch?

Yes	If only the adjustable steering column is inoperative using the memory SET switch, GO to B5 .
No	If the seats and mirrors are also inoperative using the memory SET switch, REFER to Section 501-10.

B4 CHECK THE OPERATION OF THE MEMORY SEATS, MIRRORS AND STEERING COLUMN USING AN RKE TRANSMITTER

- Check the operation of the seats, mirrors and steering column using an RKE transmitter. Refer to the Owner's Literature for additional instructions for programming a memory position to an RKE transmitter.

Do the seats and mirrors operate correctly using an RKE transmitter?

Yes	If only the adjustable steering column is inoperative using an <u>RKE</u> transmitter, GO to B5 .
No	If the seats and mirrors are also inoperative using an <u>RKE</u> transmitter, REFER to Section 501-10.

B5 CHECK FOR DSM , BCM AND SCCM DTCS

- Ignition ON.
- Enter the following diagnostic mode on the scan tool: Self Test — DSM .
- Enter the following diagnostic mode on the scan tool: Self Test — BCM .
- Enter the following diagnostic mode on the scan tool: Self-Test — SCCM .
- Record all DTCs retrieved from the DSM , BCM and SCCM .

Are any symptom or communication related DTCs present?

Yes	To diagnose the <u>DSM</u> DTCs, REFER to Section 501-10. To diagnose the <u>BCM</u> DTCs, REFER to Section 419-10. To diagnose the <u>SCCM</u> DTCs, REFER to Section 211-05.
No	INSTALL a new <u>SCCM</u> . REFER to Section 211-05. TEST the system for normal operation.

Refer to Wiring Diagrams Cell [128](#) , Adjustable Steering Column for schematic and connector information.

Normal Operation

The Steering Column Control Module (SCCM) monitors the position of the adjustable steering column using of Hall-effect type rotation sensors that are integral to the tilt and telescoping motors. Both the adjustable steering column tilt motor and telescope motor can be serviced separately from the steering column.

The SCCM sends a reference voltage along the sensor feed circuit through the Hall-effect sensor and back to the SCCM through a shared signal return circuit. The sensors provide a digital signal used by the SCCM to calculate the position of the steering column in relation to the full forward/rearward and full up/down positions. The SCCM uses this signal to store and recall memory positions, to carry out the easy entry/exit operations and to make sure the adjustable steering column tilt and telescope motors do not stall against the steering column end of travel.

If an adjustable steering column tilt or telescope motor sensor DTC sets (or if either of the sensor circuits become open), the SCCM does not recall a programmed memory position or automatically position the steering column during the easy entry/exit operations. The affected steering column motor operates in jog mode. Jog mode allows limited operation of the affected steering column motor using only the steering column control switch. When the steering column control switch is operated in jog mode, the steering column moves in the desired direction for one second, then stops regardless of the length of time the switch is actually activated. The steering column control switch must be released, then activated again in order to move the steering column for an additional second.

DTC Description	Fault Trigger Conditions
B1C33:11 — Steering Column Tilt Feedback Signal: Circuit Short To Ground	This DTC sets when the SCCM senses the adjustable steering column tilt motor feedback sensor signal is below 1.5 volts. This may be caused by a short to ground in the sensor signal circuit.
B1C33:12 — Steering Column Tilt Feedback Signal: Circuit Short To Battery	This DTC sets when the SCCM senses the adjustable steering column tilt motor feedback sensor signal is above 7 volts. This may be caused by a short to voltage in the sensor signal circuit.
B1C35:11 — Steering Column Telescope Feedback Signal: Circuit Short To Ground	This DTC sets when the SCCM senses the adjustable steering column telescope motor feedback sensor signal is below 1.5 volts. This may be caused by a short to ground in the sensor signal circuit.
B1C35:12 — Steering Column Telescope Feedback Signal: Circuit Short To Battery	This DTC sets when the SCCM senses the adjustable steering column telescope motor feedback sensor signal is above 7 volts. This may be caused by a short to voltage in the sensor signal circuit.

This pinpoint test is intended to diagnose the following:

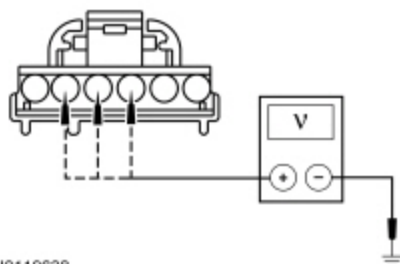
- Wiring, terminals or connectors
- Adjustable steering column tilt motor
- Adjustable steering column telescopic motor
- SCCM

PINPOINT TEST C : THE ADJUSTABLE STEERING COLUMN OPERATES IN ONE SECOND INTERVALS WHEN USING THE STEERING COLUMN CONTROL SWITCH (JOG MODE)

NOTICE: Use the Flex Probe Kit for all test connections to prevent damage to the wiring terminals. Do not use standard multi-meter probes.

C1 CHECK THE TILT/TELESCOPE SENSOR CIRCUITS FOR A SHORT TO VOLTAGE

- Ignition OFF.
- Disconnect: SCCM [C2414C](#) .
- Disconnect: Adjustable Steering Column Tilt Motor [C2600](#).
- Disconnect: Adjustable Steering Column Telescope Motor [C2601](#).
- Ignition ON.
- Measure the voltage between ground and:
 - SCCM [C2414C](#) Pin 3, circuit RPP06 (GN), harness side.
 - SCCM [C2414C](#) Pin 4, circuit VPP08 (WH), harness side.
 - SCCM [C2414C](#) Pin 5, circuit VPP17 (VT/BN), harness side.

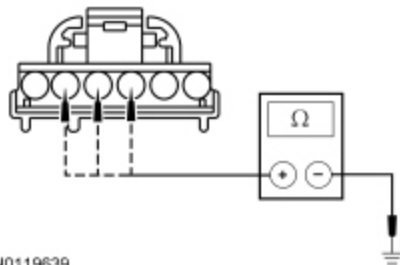


Is any voltage present?

Yes	REPAIR the affected circuit(s). CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.
No	GO to C2 .

C2 CHECK THE TILT/TELESCOPE SENSOR CIRCUITS FOR A SHORT TO GROUND

- Ignition OFF.
- Measure the resistance between ground and:
 - SCCM [C2414C](#) Pin 3, circuit RPP06 (GN), harness side.
 - SCCM [C2414C](#) Pin 4, circuit VPP08 (WH), harness side.
 - SCCM [C2414C](#) Pin 5, circuit VPP17 (VT/BN), harness side.



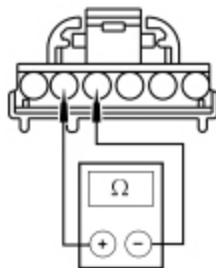
Is the resistance greater than 10,000 ohms?

Yes	GO to C3 .
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No	REPAIR the affected circuit(s). CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.
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C3 CHECK THE TILT/TELESCOPE SENSOR CIRCUITS FOR A SHORT TOGETHER

- Measure the resistance between SCCM [C2414C](#) Pin 4, circuit VPP08 (WH), harness side and SCCM [C2414C](#) Pin 5, circuit VPP17 (VT/BN), harness side.



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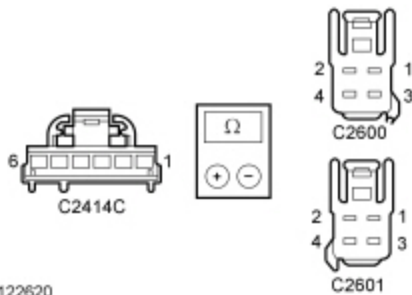
Is the resistance greater than 10,000 ohms?

Yes	GO to C4 .
No	REPAIR the affected circuit(s). CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

C4 CHECK THE TILT/TELESCOPE SENSOR CIRCUITS FOR AN OPEN

- Measure the resistance between SCCM [C2414C](#), harness side and adjustable steering column tilt motor [C2600](#); and between SCCM [C2414C](#), harness side and adjustable steering column telescope motor [C2601](#), harness side using the following chart:

SCCM	Circuit	Adjustable Steering Column Tilt/Telescope Motor
C2414C Pin 3	RPP06 (GN)	C2600 Pin 3
C2414C Pin 3	RPP06 (GN)	C2601 Pin 3
C2414C Pin 4	VPP08 (WH)	C2600 Pin 4
C2414C Pin 5	VPP17 (VT/BN)	C2601 Pin 4



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Are the resistances less than 5 ohms?

Yes	GO to C5 .
No	REPAIR the affected circuit(s). CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.

C5 CHECK THE SCCM CONNECTOR

- Check SCCM connector [C2414C](#) for:
 - corrosion
 - damaged pins
 - pushed-out pins
- Connect: SCCM [C2414C](#) .
- Connect: Adjustable Steering Column Tilt Motor [C2600](#).
- Connect: Adjustable Steering Column Telescope Motor [C2601](#).
- Ignition ON.
- Make sure the connectors are seated correctly then operate the system and verify the concern is still present.

Is the concern still present?

Yes	If the adjustable steering column does not operate correctly when the steering column control switch is in the UP/DOWN (tilt) position(s), INSTALL a new adjustable steering column tilt motor. REFER to Steering Column Motor — Tilt in this section. CLEAR the DTCs. REPEAT the self-test. If the concern is still present, INSTALL a new SCCM . REFER to Section 211-05. TEST the system for normal operation. If the adjustable steering column does not operate correctly when the steering column control switch is in the FORWARD/REARWARD (telescope) position(s), INSTALL a new adjustable steering column telescopic motor. REFER to Steering Column Motor — Telescopic in this section. CLEAR the DTCs. REPEAT the self-test. If the concern is still present, INSTALL a new SCCM . REFER to Section 211-05. TEST the system for normal operation.
No	The system is operating correctly at this time. Concern may have been caused by a loose or corroded connector. CLEAR the DTCs. REPEAT the self-test. TEST the system for normal operation.